MILITARY SPECIFICATION SHEET

INTERCOMMUNICATION SETS AN/AIC-14 AND AN/AIC-14A

PASSIVE INTERPHONE CONTROL PANEL

C-2647A/AIC-14

Inactive for new design after 10 August 1984
For new design use MIL-I-22352/23(AS).

This specification sheet is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the unit described herein shall consist of this document and the latest issue of specification MIL-I-22352(AS).

1. SCOPE

1.1 Scope. This specification sheet provides performance and physical characteristics for the passive Interphone Control panel C-2647A/AIC-14, hereinafter referred to as C-2647A when used for ICS operation.

1.2 Military Part Number. The military part number shall be M22352/13-1

2. APPLICABLE DOCUMENTS

Refer to 20 of MIL-I-22352(AS) Appendix A for a listing of applicable documents. The requirements specified herein shall have precedence over all referenced documents. The contract and its attachments shall have precedence over any specification or reference document.

3. REQUIREMENTS

3.1 Item definition. The passive Interphone Control panel C-2647A shall provide:

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(1) Interconnection with other interphone stations. One through eleven other interphone stations may be selected at the operators option for outgoing interphone communications.

(2) Signal processing consisting of summing of eleven incoming interphone signals.

3.1.1 Item diagram. Passive Interphone Control panel C-2647A shall provide the functions represented schematically in accordance with the diagram of Figure 1.

3.1.2 Interface definition. The passive Interphone Control Panel input and output signals will normally exhibit the characteristics presented in Table I.

TABLE I. Interface signal list.

Signal Name	Characteristics
Inputs:	
Interphone No. 1 (INTPH NO. 1)	Voice range audio (300 to 6,000 Hz) signal ranging in amplitude from 0 to 1 Vac.
Interphone No. 2 through Interphone No. 11 (INTPH NO. X)	Same as Interphone No. 1
Interphone (INTPH)	Voice range audio (300 to 6,000 Hz) signal ranging in amplitude from 0 to 1 Vac.
CALL Emergency Override (CALL)	Relay logic signal: 1) Open 2) Short to ground
Interphone Key (INTPH KEY)	Relay logic signal: 1) Open 2) Short to ground
Panel Lights (LIGHTS)	0 to + 28Vdc or 0 to 6V, 400 Hz
Power (PWR)	+27.5 ± 0.5 Vdc
Outputs:	
Interphone (INTPH)	Voice range audio (300 to 6,000 Hz) signal ranging in amplitude from 0 to 1 Vac. No dc component normally present.

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TABLE I. Interface signal list-continued

Signal Name	Characteristics
Common Audio (COMMON AUDIO)	Voice range audio (300 to 6,000 Hz) signal ranging in amplitude from 0 to 1 Vac.
Interphone No. 1 (INTPH NO. 1)	Voice range audio (300 to 6,000 Hz) signal ranging in amplitude from 0 to 1 Vac. No dc component normally present.
Interphone No. 2 through Interphone No. 12 (INTPH NO. X)	Same as Interphone No. 1
Grounds:	
Power (PWR GND)	Power return. See 3.4.4.2 of MIL-I-22352(AS).
System (SYS GND)	Signal return. See 3.4.4.3 of MIL-I-22352(AS).

3.1.3 Controls. The passive Interphone Control panel shall provide the controls listed in Table II.

3.1.4 Indicators. The passive Interphone Control panel shall provide three front panel lamps to enhance night vision.

3.2 Characteristics

3.2.1 Performance characteristics

3.2.1.1 Input characteristics

3.2.1.1.1 Interphone NO. 1 through Interphone No. 11 (INTPH NO. X). The INTPH NO. 1 through INTPH NO. 11 input characteristics shall be as follows:

3.2.1.1.1.1 Signal Frequency. The INTPH NC. X input shall accept drive signals which range from 300 to 6,000 Hz.

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TABLE II. C-2647A controls list.

Title	Function
ICS	The Interphone Control panel shall provide a series of twelve two position toggle switches as defined herein. The switches shall provide the operator at each Interphone Station with the capability to select the following functions: SELECTED - In the up position the operator will receive the interphone audio of the Interphone Station associated with the switch actuated. OFF - In the down position the system outputs are disabled.
ALL	CALL - In the down position the Override Relay shall be actuated by a LO on the CALL input but unaffected by the state of the INTPH KEY input. ALL - In the up position the Override Relay shall be actuated by a LO on either the CALL or the INTPH KEY input.

3.2.1.1.1.2 Signal Amplitude. The INTPH NO. X inputs shall accept drive signals which range in amplitude from 0 to 10 Vac.
3.2.1.1.1.3 Input Impedance. The INTPH inputs shall each present a 62,000 ohm ±10 percent impedance to the incoming signals.

3.2.1.1.2 Common Audio (COMMON AUDIO). The COMMON AUDIO line shall act as an output as well as an input. See 3.2.1.2.3 for output characteristics.

The COMMON AUDIO shall be routed via the Override Relay to the INTPH output except when the Override Relay is energized by a CALL input or alternatively by a LO at the INTPH KEY input when the ALL switch is set to the ALL position.

3.2.1.1.3 Interphone (INTPH).

3.2.1.1.3.1 Signal Frequency. The INTPH input shall accept drive signals which range from 300 to 6,000 Hz.

3.2.1.1.3.2 Signal Amplitude. The INTPH input shall accept drive signals which range in amplitude from 0 to 10 Vac.

3.2.1.1.4 CALL (CALL). The CALL input shall accept a relay drive logic signal having two states: 1) open and 2) short to ground. The CALL input shall be applied to the Override Relay which connects the INTPH input of the COMMON AUDIO output of 3.2.1.2.3.

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3.2.1.1.5 Key External (KEY EXT). The KEY EXT input shall accept a relay drive logic signal having two states: 1) open and 2) short to ground. The KEY EXT signal shall be applied to the ALL switch.

3.2.1.1.6 Power (PWR). The C-2647A shall accept and operate within the constraints of this specification when supplied with +27.5 ±0.5 Vdc electrical power as shown in the diagram of Figure 1. The maximum load presented by the C-2647A shall not exceed 100 milliamperes.

3.2.1.1.7 Panel Lights (LIGHTS). The C-2647A shall be capable of accepting +28 Vdc or 6 Vac at 400 Hz power to illuminate the three front panel lamps. The panel lamps used shall be selected from Table III.

TABLE III. Front panel bulbs.

Nomenclature	Power
MS25237A-327 MS25237A-328	+28 Vdc 6 Vac

3.2.1.2 Output Characteristics. The C-2647A shall have the following output characteristics:

3.2.1.2.1 Interphone NO. 1 through Interphone No. 11 (INTPH NO. X). It shall be possible to select any combination of the INTPH NO. 1 through INTPH NO. 11 outputs to have active signals simultaneously.

3.2.1.2.2 Mixed Interphone (MIXED INTPH). The MIXED INTPH output shall be the sum of operator selected INTPH NO. 1 through INTPH NO. 12 inputs. Selection is made by setting the desired ICS switches to the SELECTED (up) position. The INTPH output amplitude shall be 0 ±4 dB with respect to a 0.03 Vac, 1,000 Hz input at INTPH NO. 1 through INTPH NO. 11.

The MIXED INTPH output shall be 0 ±4 dB with respect to a 0.03 Vac, 1,000 Hz at the COMMON AUDIO input when the Override Relay is not activated.

3.2.1.2.3 Common Audio (COMMON AUDIO). The COMMON AUDIO shall be connected to the INTPH input of Section 3.2.1.1.1 when the Override Relay is energized.

3.2.1.2.4 Panel lamps (LIGHTS). The panel lamps shall be edge light type having a red lens in accordance with MIL-L-25467. The C-2647A shall be capable of operating with lamp power ranging up to +28 Vdc or 6 Vac at 400 Hz indefinitely without damage.

The unit shall be capable of operating with either ac or dc lighting power source as permitted in MIL-C-6781. When ac power is used for panel lighting self susceptibility to the 400 Hz power source shall induce not more than 5 millivolts across any properly terminated output with the input circuits shorted.

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3.2.2 Physical characteristics.

3.2.2.1 Dimensions. The C-2647A passive Interphone Control panel shall consist of a unit having the dimensions of Figure 2. All outside dimensions shall be within ±0.03 inches of the nominal value shown.

3.2.2.2 Weight. The C-2647A shall weigh no more than 1.75 pounds.

3.3 DESIGN AND CONSTRUCTION

The design and construction of the C-2647A passive Interphone Control panel shall be in accordance with the requirements of 3.3 of MIL-I-22352(AS).

3.3.1 Plastic lighting plates. Plastic lighting plates shall meet the requirements of MIL-P-7788 and Figure 2 of this specification. Markings shown on Figure 2 shall conform with the appropriate application requirements of MS25492. A listing of lighting plates for the C-2647A is contained within Supplement 1 to this specification. If the appropriate configuration drawings are not specified in the contract, this information will be supplied by the acquiring activity upon application by the contractor.

3.3.2 Front panel lighting. The front panel lamps shall be installed using mounting hardware having red filters in accordance with MIL-L-25467 to enhance night vision. The panel lamps shall be mounted in accordance with the outline drawing Figure 2. The lamp drive circuitry shall be capable of accommodating input power ranging up to 28 Vdc or 6 V, 400 Hz ac.

3.3.3 Electrical Connections. A shorting plug designated P702 shall be provided internal to the C-2647A to facilitate use as a Radio Receiver Selector panel or as an ICS selector panel. Plug P702 shall be supplied installed in the C-2647A when delivered. Removal of the plug converts operation of the C-2647A from ICS Selector to Receiver Selector.

4. QUALITY ASSURANCE PROVISIONS

See MIL-I-22352(AS).

4.1 Responsibility for tests and inspections. See MIL-I-22352(AS).

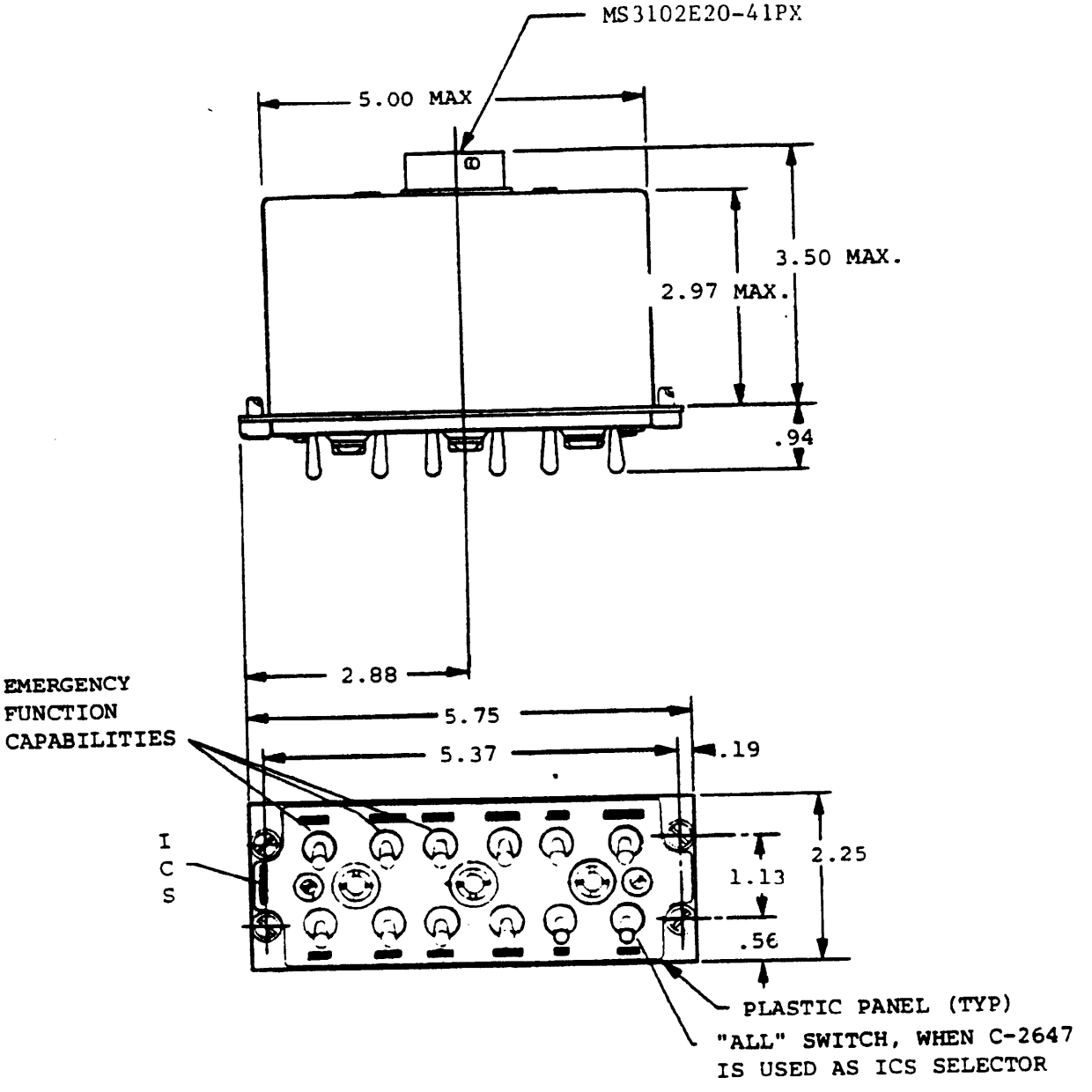
4.2 Test plan. A detailed test plan will be submitted by the contractor for approval by the acquiring activity which details the tests to be performed in order to verify compliance with the performance characteristics specified in 3.2.1. The test plan will consider all of the inspections delineated in 4.2 of MIL-I-22352(AS).

4.3 Test procedures. Detailed test procedures will be submitted by the contractor for approval by the acquiring activity which will satisfy the requirements of the test plan of 4.2.

Preparing Activity:
Navy - AS

Project No. 5831-N042-13

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1/ All dimensions are in inches. Tolerances on dimensions are ±0.03 inches unless otherwise specified.

FIGURE 2. C-2647A outline drawing.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-I-22352/13(AS)	INTERCOMMUNICATION SETS AN/AIC-14 AND AN/AIC-14A PASSIVE INTERPHONE CONTROL PANEL C-2647A/AIC-14

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
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5 PROBLEM AREAS

a. Paragraph Number and Wording.

b. Recommended Wording.

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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